Building AI Agents
To Conduct Interview

AI Job Interview Bot

Question 1 of 2

What is the difference between a list and a tuple in Python, and when would you choose one over the other?

Your Answer

List are defined using square bracket[], where as tuple are defined using round bracket (). List elements can be modified. Tuple elements cannot be modified. Tuples are immutable. List has lot many methods available for operation, whereas tuples has 2 or 3 methods available

Submit Answer

Question 2 of 2

What is the purpose of the `self` parameter in instance methods within a Python class?

Feedback

Question asked: What is the purpose of the `self` parameter in instance methods within a Python class?

Your Answer: It allows an instance method to access the object's attributes and other methods.

Score: 7 out of 10

Strengths:

- The candidate correctly identified that `self` allows access to the object's attributes and methods, demonstrating an understanding of instance methods.

Weaknesses:

- The explanation lacks detail on how `self` differentiates between instance variables and methods, and it does not mention that `self` must be the first parameter in instance methods. Additionally, a brief mention of its role in referencing the current instance could enhance clarity and completeness.

🎉 Interview Completed!

Final Score: 7.00 / 10

Thank you for completing the AI interview.

Restart Interview

1. What is an AI Agent?

An **AI Agent** is a system that can:

1. Receive an input
2. Think using prompts or rules
3. Take actions or produce responses

Simple Flow:

User Input → AI Prompt → AI Reasoning → Output → Next Action

Example:
User: Help me prepare for a job interview

AI Agent would
1. Ask questions from user
2. Evaluate answers
3. Give feedback

This is exactly what we will build today.

Part 1: System Prompts (Setting the "Brain" of AI)

What is a System Prompt?

A **system prompt defines the personality and behavior of the AI**.

It tells the AI:

- Who it is and what role it should play
- How it should behave
- What rules to follow

Example Without System Prompt

Ask me interview questions

Example With System Prompt

You are an experienced technical HR interviewer.

You are interviewing a candidate for a Python Developer role.

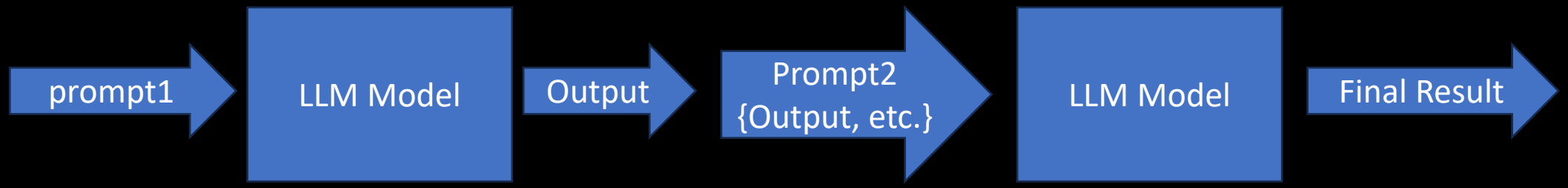
Your behavior:
- Ask only ONE question at a time
- The question must be conceptual (not coding)
- The question should test understanding of Python fundamentals
- Keep the question clear and concise.
- Do not provide explanations or answers

Part 2: Prompt Chaining

What is Prompt Chaining?

Prompt chaining means:
Output of one prompt becomes the input to another prompt

Flow:
Prompt 1 → LLM → Output
Output → Prompt 2
Prompt 2 → LLM → Final Result
This creates **multi-step reasoning**.



Example

Suppose we want to:

1. Ask interview question
2. Evaluate answer
3. Give feedback

Step 1: Ask Question

Prompt1:

```
You are a Python interviewer.
Ask one interview question.
```

Output:

```
What is the difference between a list and a tuple in Python?
```

Here user answers the question.

```
List are mutable. Tuples are immutable
```

Step 2: Evaluate Answer

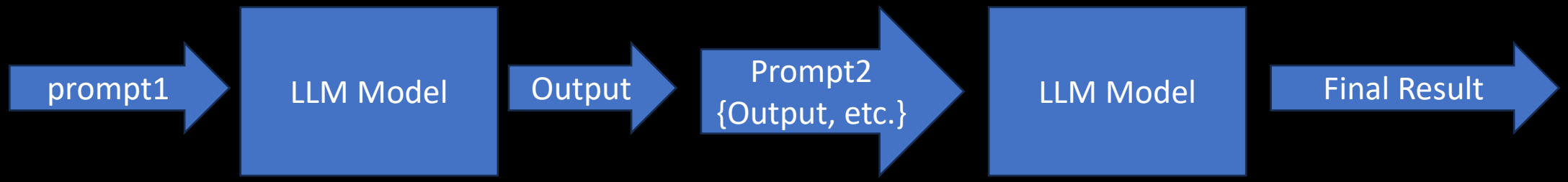
Now we feed the question generated and the answer.

Prompt2:

```
Evaluate the following interview answer.

Question:
What is the difference between list and tuple?

Answer:
List are mutable. Tuples are immutable


Give feedback and score out of 10.
```

This was the output from prompt1

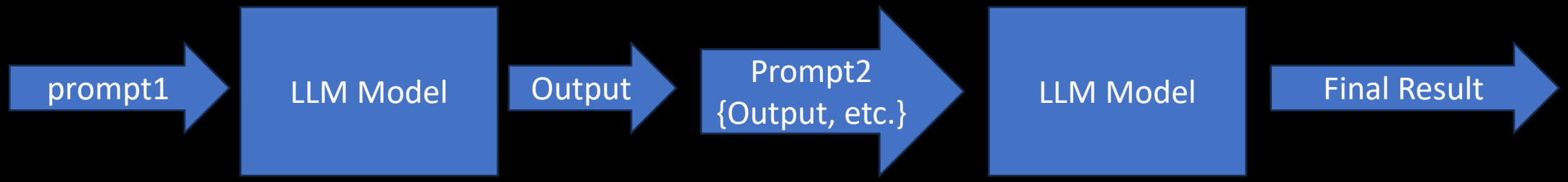
This is the users reply



Step 3: Give feedback

Final result

```
You are right about the core concept. Your answer
could have been better if you had provided examples.

You scored 7 out of 10
```

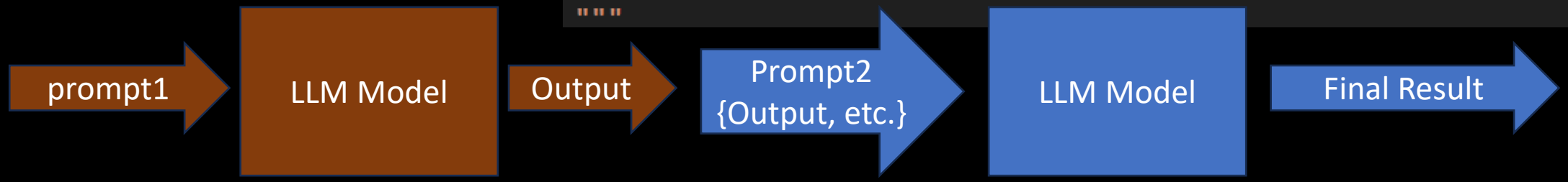


If you are using python then install following 2 packages:

python -m pip install openai
python -m pip install python-dotenv

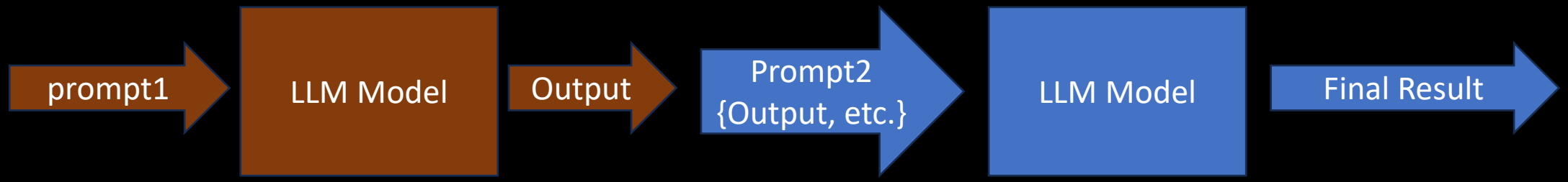
step1_generate_question.py

```python
from openai import OpenAI
import os
from dotenv import load_dotenv

load_dotenv()
OPENAI_API_KEY = os.getenv('OPENAI_API_KEY')
client = OpenAI(api_key=OPENAI_API_KEY)

# system prompt defines the behavior, personality, and rules of the AI.
system_prompt = """
You are an experienced technical HR interviewer.

You are interviewing a candidate for a Python Developer role.

Your behavior:
- Ask only ONE question at a time
- The question must be conceptual (not coding)
- The question should test understanding of Python fundamentals
- Keep the question clear and concise
- Do not provide explanations or answers
"""
```



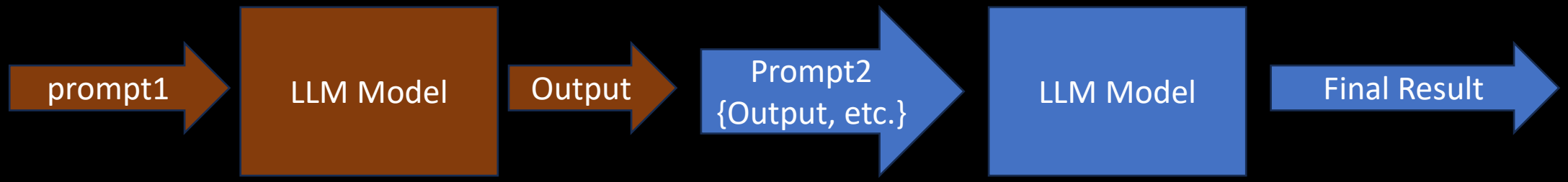
step1_generate_question.py

```python
question_prompt = """Generate one Python interview question.

Constraints:
- The question should be suitable for an entry to mid-level Python developer
"""

response = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[
        {"role": "system", "content": system_prompt},
        {"role": "user", "content": question_prompt}
    ]
)

print(response)
question = response.choices[0].message.content
print(question)
```



step1_generate_question.py

```
# ChatCompletion(id='chatcmpl-DI83j1iHD3IpdFzYydZkFDgmNZNwq', choices=[Choice(finish_reason='stop', index=0,
logprobs=None, message=ChatCompletionMessage(content='What is the difference between a list and a tuple in Python,
and when would you choose to use one over the other?', refusal=None, role='assistant', annotations=[], audio=None,
function_call=None, tool_calls=None))], created=1773213255, model='gpt-4o-mini-2024-07-18', object='chat.
completion', service_tier='default', system_fingerprint='fp_687f94b198', usage=CompletionUsage
(completion_tokens=25, prompt_tokens=102, total_tokens=127, completion_tokens_details=CompletionTokensDetails
(accepted_prediction_tokens=0, audio_tokens=0, reasoning_tokens=0, rejected_prediction_tokens=0),
prompt_tokens_details=PromptTokensDetails(audio_tokens=0, cached_tokens=0)))
# What is the difference between a list and a tuple in Python, and when would you choose to use one over the other?
```



For
step2_xxx.py
step3_xxx.py
step4_xxx.py

Look at the code